

SuperCart

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Executive Summary-SuperKart Revenue Forecasting System

Business Problem & Context:

SuperKart is a rapidly expanding retail chain operating across Tier-1 and Tier-2 cities. Currently, the organization faces significant challenges in regional sales strategy and inventory management due to a lack of precise outlet-level revenue forecasts. Without accurate predictive insights, SuperKart risks misallocating resources, overstocking perishables, and missing critical revenue opportunities in its diverse store network

Strategic Objective :

The primary objective was to develop and deploy a scalable, high-precision forecasting solution capable of predicting upcoming quarterly sales revenue. By leveraging historical sales data and key external drivers—such as Product MRP, Store Age, and Location Type—this system enables data-driven decision-making to optimize supply chain efficiency and reduce operational costs.

The Winner Model : Base Random Forest

After a rigorous comparative analysis of multiple algorithms (including XGBoost, Hyper parameter tuning for Random Forest and XGBoost Regressor), the **Base Random Forest** model emerged as the definitive "**Winner**" for production.

- **Top-Tier Performance:** Achieved the highest predictive accuracy with a **Test R^2 of 0.9232**, outperforming even the tuned versions of complex models like XGBoost.
- **Robust Generalization:** The minimal gap between the **Training R^2 (0.9903)** and **Test R^2 (0.9232)** demonstrates the model's exceptional ability to generalize to unseen data without overfitting.
- **Reliability:** The model recorded the lowest **Root Mean Square Error (RMSE) of 296.43** on the test set, making it the most reliable tool for avoiding costly inventory errors.
- **Key Drivers:** The ensemble effectively captured non-linear relationships, identifying **Product MRP** and **Product Weight** as the most significant drivers of store revenue.

Executive Summary-SuperKart Revenue Forecasting System

Pre-processing & Feature Engineering Excellence:

The model's success is anchored in a robust preprocessing pipeline that transformed raw data into high-value signals:

- **Data Refinement:** Standardized inconsistent labeling (e.g., merging "reg" into "Regular" sugar content) and simplified 16 product types into **Perishable vs. Non-Perishable** categories to reduce model noise.
- **Derived Intelligence:** Engineered a **"Store Age"** feature, providing the model with a clear metric for store maturity and experience.
- **Pipeline Integrity:** Utilized a **Scikit-learn Pipeline** to unify scaling and encoding, which eliminated the risk of data leakage and ensured consistent performance during real-time inference.

Deployment & Accessibility (Hugging Face):

To ensure the model provides immediate business value, it has been deployed using a modern, **decoupled architecture** on the **Hugging Face** platform:

- **Backend (The "Brain"):** A Flask-based API serves as the inference engine, containerized via **Docker** to ensure the model performs consistently in any environment. Backend API : <https://singh00in-SuperKartNewBackend.hf.space/v1/predict>
- **Frontend (The "Face"):** A user-friendly **Streamlit** application allows SuperKart's internal teams to input store parameters and receive instant revenue forecasts via a web-accessible URL. <https://singh00in-SuperKartFrontend.hf.space>
- **Scalability:** This architecture allows the forecasting engine to be integrated into broader enterprise systems while remaining independently scalable.

Executive Summary-SuperKart Revenue Forecasting System

Actionable Insights:

- ✓ **MRP as the Primary Revenue Driver:** Product_MRP shows the strongest positive correlation (0.79) with total sales, indicating that higher-priced items significantly drive overall revenue. Strategic pricing or a focus on premium items in high-performing stores could further boost total turnover.
- ✓ **Targeting Tier-2 Market Dominance:** Tier-2 cities are the primary operational focus and revenue engine, generating over 20 million in total sales—more than double the revenue of Tier-1 and Tier-3 cities combined. This identifies Tier-2 as the most critical market for expansion and resource allocation.
- ✓ **Optimizing for Perishables:** "Fruits and Vegetables" is the leading product category by both volume and revenue, highlighting a critical need for high-precision forecasting to manage the logistics of perishable inventory effectively.
- ✓ **Store-Level Performance Variance:** While Store OUT004 has the highest transaction volume, Store OUT003 exhibits the highest median sales per product, suggesting a more "high-value" customer base at that specific location.
- ✓ **Inefficiency of Shelf Space:** Product_Allocated_Area has a near-zero correlation (0.00) with sales revenue, meaning that assigning more physical floor space to a product does not statistically guarantee higher sales

Strategic Business Recommendations :

- ✓ **Deploy the Base Random Forest Model:** With a **Test R^2 of 0.9232** and the lowest **RMSE (296.43)**, the Base Random Forest is the most reliable model for production. It captures complex, non-linear relationships much more effectively than the tuned versions or other regressors
- ✓ **Refine Inventory Based on Store Type:** Supermarket Type2 is significantly more profitable than Food Marts or Departmental Stores. SuperKart should prioritize the Supermarket Type2 format for new store openings to maximize potential ROI
- ✓ **Merge and Standardize Data Labels:** Future data collection should address labeling inconsistencies, specifically merging the "reg" sugar content category into "Regular" to improve data quality and future model training.
- ✓ **Leverage High-Value "Superstars":** The presence of significant positive sales outliers (up to 8,000) suggests that certain "star" products or stores perform exceptionally well. Identifying the shared characteristics of these outliers could help replicate their success across the broader network.

Business Problem Overview

Business Context :

Sales forecasting is a critical business function that predicts future revenue based on historical sales, industry trends, and the current sales pipeline. **Accurate forecasts** enable organizations to estimate weekly, monthly, quarterly, and annual sales, **providing actionable insights across multiple business verticals**.

For retail businesses, **effective sales forecasting** drives operational efficiency by guiding inventory management, regional sales strategies, and supply chain planning. It helps **reduce sales pipeline** risks, improves decision-making, optimizes territory coverage, and establishes performance benchmarks to track trends over time.

Problem Statement :

SuperKart, a rapidly expanding retail chain with supermarkets and food marts across tier-1 and tier-2 cities, faces challenges in managing inventory and aligning regional sales strategies due to the lack of precise sales forecasts. Without an accurate prediction of outlet-level revenue, the company risks overstocking or understocking products, misallocating resources, and missing potential revenue opportunities

Objective :

To address these challenges, SuperKart **aims to accurately forecast** the sales revenue of its outlets for the upcoming quarter. The company has partnered with a data science firm to:

1. Develop a **predictive model** leveraging historical sales data and relevant external factors.
2. Deploy a **scalable forecasting solution** that integrates with SuperKart's decision-making systems.
3. Enable data-driven decisions across its network of stores, **optimizing inventory management** and **regional sales planning**.

Expected Impact :

- **Reduce** inventory-related costs and stockouts.
- Make informed, region-specific sales and marketing decisions.
- Streamline supply chain operations and procurement planning.
- Improve **overall organizational efficiency** and strategic planning capabilities.

Solution Approach

1. Data Analysis & Data Background Verification : Understanding the ground truth of the data

- Understand the Shape of the Data
- Verifications of the Data Types
- Data Completeness Check : Missing Value verifications
- Data Consistency Check : Verification of duplicates , units , or incorrect data
- Check the statistical summary of the data

2. Exploratory Data Analysis (EDA) : To understand various pattern and Business Driver in data sets

Univariate Analysis: Understand the distribution of

- Product_Weight ,Product_Allocated_Area ,Product_MRP ,Product_Store_Sales_Total ,Product_Sugar_Content ,Product_Type
- Store_ID,Store_Type , Store_Location_City_Type ,Store_Size

Bivariate Analysis

- Correlation matrix
- Check the distribution of our target variable i.e Product_Store_Sales_Total with the numeric columns
- Understand which product type the company is generating most of the revenue
- Check which type of stores and locations the revenue generation is more
- Check the distribution of our target variable i.e Product_Store_Sales_Total with the other categorical columns
- Find out whether there is some relationship between the weight of the product and its sugar content
- Analyze the sugar content of different product types
- Find out how many items of each product type has been sold in each of the stores
- Different product types have different prices. Let's analyze the trend.
- Find out how the Product_MRP varies with the different stores
- Delve deeper and do a detailed analysis of each of the stores.

Solution Approach

3. Data Preprocessing & Pipeline: This is where raw numbers transform into signals for the model.

- Replacing the values in the Product_Sugar_Content column
- Exploring Patterns in Product_IDs , and extarctation of first Two Char and removing Product_ID column
- Calculation of the Store Age as a store which has been in the business for a long duration is more trustworthy than the newly established ones.
- Grouping Product Types into Perishables and Non-Perishables.
- Outlier Check

4. Data Preparation

- Target to forecast the **Product_Store_Sales_Total**.
- Drop unnecessary columns and encode the categorical features.
- Split the data into training and testing sets with the **ratio of 70:30** to evaluate the model's performance on unseen data.

5. Preprocessing Pipeline: Use Scikit-learn Pipelines to handle scaling and encoding consistently during both training and inference.

6. Model Building : My strategy is to move from simple to complex to ensure the added complexity provides actual value.

- Define functions for Model Evaluation , **R-square - Coefficient** of determination is used to evaluate the performance of a regression model. It is the amount of the variation in the output dependent attribute which is predictable from the input independent variables.
- Create Base Model for (**Decision Tree, Bagging Regressor , Random Forest , AdaBoost Regressor , GB Regressor and XGB Regressor**) and take a note for the Performance Metrics
- As next step identify the appropriate parameters and tune above model , use **GridSearchCv for hyperparameter tuning and r_2 score** to optimize the model
- Compare their performance for Training and Test data Sets based on the **R-square** and finalize the Best model for SuperKart Production
- Serialize best model and save it to proceed for deployment

Solution Approach

7. Deployment Architecture (Decoupled Approach)

Strategy : Decouple the frontend and backend is excellent for scalability. Here is how the flow looks:

1. Backend (The "Brain")

- **Flask API:** Acts as the inference engine. It loads the saved model (**.joblib**) and processes **JSON** requests.
- **Exposing API End**
- **Containerization:** The Dockerfile ensures that the "Brain" works the same on local machine as it does on Hugging Face.
- This backend handles the core prediction logic, applying the same data preprocessing steps (imputation, scaling, and encoding) used during model training to ensure consistency and accuracy. It returns predictions as JSON responses, facilitating easy integration with the frontend.
- Deploy this Flask app, the serialized model, and the requirements.txt file to a Hugging Face Space using a Dockerfile. This makes the prediction service publicly accessible via a unique URL.

2. Frontend (The "Face")

- **Streamlit:** Designed for SuperKart's internal teams to input store parameters and get an instant **revenue forecast**.
- **Communication:** Streamlit uses the requests library to **POST data to the Flask API endpoint** and receive the prediction.
- Similar to the backend, we'll deploy the Streamlit app to a Hugging Face Space using a Dockerfile with its own requirements.txt for managing dependencies.

Data Overview

- **Shape of the Data set** : There are **8763** rows and **12** columns.
- **Data Types Column of the DataSet** :

```
RangeIndex: 8763 entries, 0 to 8762
Data columns (total 12 columns):
#   Column                                Non-Null Count  Dtype
---  ---                                ---
0   Product_Id                            8763 non-null   object
1   Product_Weight                        8763 non-null   float64
2   Product_Sugar_Content                8763 non-null   object
3   Product_Allocated_Area               8763 non-null   float64
4   Product_Type                         8763 non-null   object
5   Product_MRP                          8763 non-null   float64
6   Store_Id                             8763 non-null   object
7   Store_Establishment_Year             8763 non-null   int64
8   Store_Size                           8763 non-null   object
9   Store_Location_City_Type             8763 non-null   object
10  Store_Type                           8763 non-null   object
11  Product_Store_Sales_Total            8763 non-null   float64
dtypes: float64(4), int64(1), object(7)
```

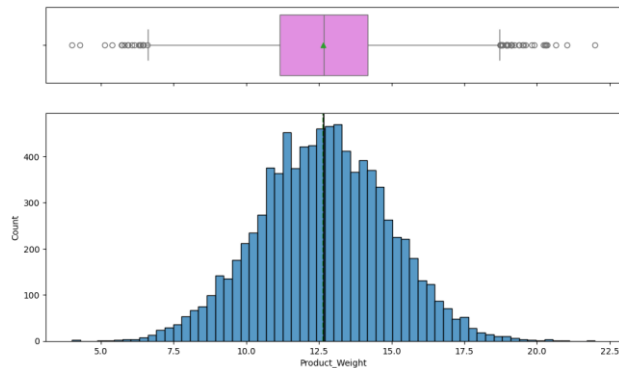
- **Duplicate Value:**
The dataset does not contain any duplicate value
- **Missing Value :**
The dataset does not contain any missing value

Statistical summary of the data:

- **Tier 2 Dominance:** Geographical data is heavily skewed toward Tier 2 cities, accounting for **6,262** out of **8,763** entries.
- **Store Concentration:** Store **OUT004** and **Supermarket Type2** dominate the dataset, suggesting a potential data imbalance or a primary flagship location.
- **High Sales Variance:** Sales totals range from **33 to 8,000**, indicating significant volatility and the presence of high-performing products.
- **Perishable Focus: Fruits and Vegetables** is the leading category, highlighting the need for high-precision forecasting for perishable inventory.
- **Area Allocation: Product_Allocated_Area** varies significantly, peaking at nearly 30% for certain items, which likely acts as a key driver of sales revenue

EDA Results- Univariate Analysis

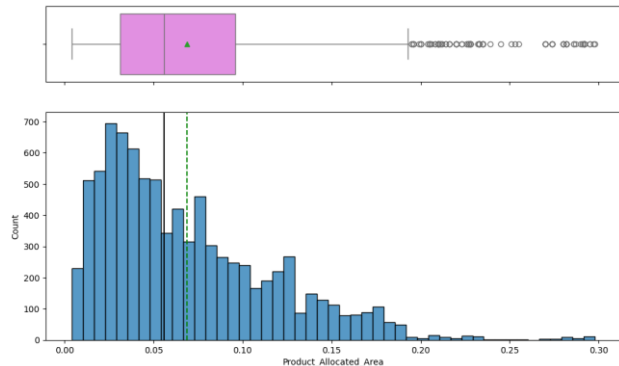
Product_Weight



Key Observations :

- **Normal Distribution:** The product weights follow a near-perfect Gaussian distribution centered around a mean of **12.65**, suggesting a well-balanced variety of product masses.
- **Low Outlier Impact:** While the boxplot identifies a few outliers at the extreme low (**4.0**) and high (**22.0**) ends, they represent a negligible fraction of the 8,763 entries.
- **Data Consistency:** The mean (**12.65**) and median (**12.66**) are almost identical, indicating highly symmetrical data with no significant skewness to address during preprocessing.

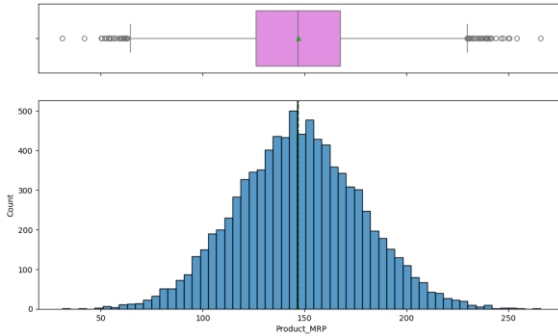
Product_Allocated_Area



- **Right-Skewed Space Allocation:** _Area is **Product_Allocated** heavily right-skewed with a mean of **0.068**, showing most products occupy small spaces while a few outliers reach up to **0.298**.
- **Concentrated Geographic Footprint:** The data shows a strong regional bias, with **71.4%** of entries (6,262 out of 8,763) originating from **Tier 2** cities.

EDA Results- Univariate Analysis

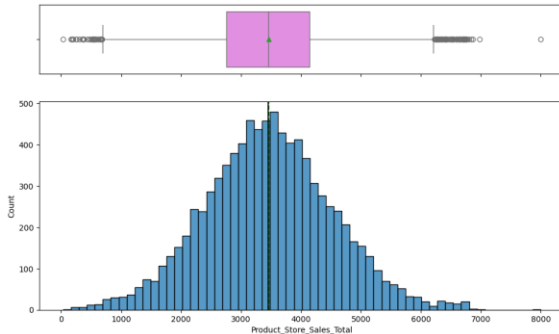
Product_MRP



Key Observations :

- **Symmetrical Distribution:** The histogram shows a nearly perfect bell-shaped normal distribution, centered around a mean of **147.03**.
- **High Density Mid-Range:** The peak frequency occurs between **140 and 160**, where the majority of products are priced.
- **Presence of Price Outliers:** The boxplot identifies several outliers at both the lower end (**near 31**) and upper end (**near 266**) of the pricing spectrum.

Product_Store_Sales_Total



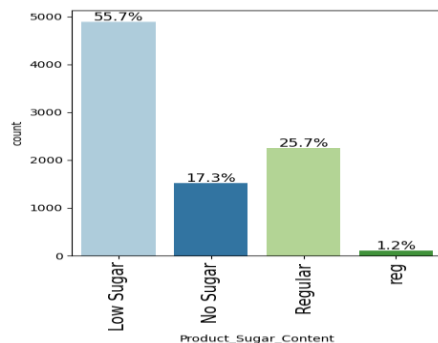
Symmetrical Distribution Profile: The **Product_Store_Sales_Total** follows a near-perfect normal distribution, with the mean (3464.00) and median (3452.34) almost perfectly aligned.

High-Value Sales Outliers: The boxplot indicates significant positive outliers extending up to a maximum value of **8000.0**, well beyond the upper whisker.

Concentrated Revenue Range: The majority of store sales are densely clustered between the **25th percentile (2761.72)** and the **75th percentile (4145.17)**.

EDA Results- Univariate Analysis

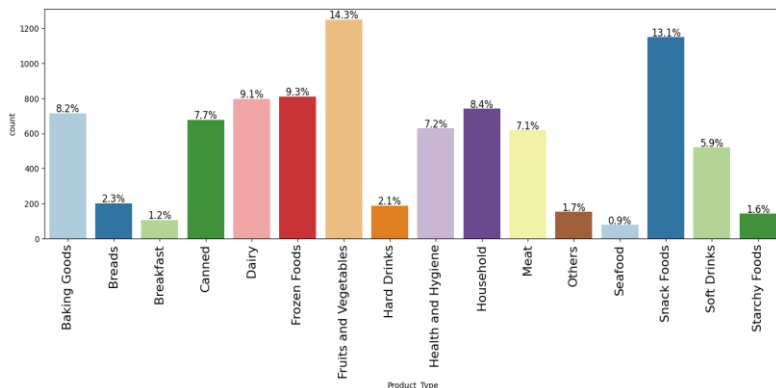
Product_Sugar_Content



Key Observations :

- **Dominant Consumer Preference: Low Sugar** products are the primary category, representing over half of the dataset at **55.7%**.
- **Secondary Market Share: Regular** sugar content items hold the second-largest share at **25.7%**, nearly double the **No Sugar** category at **17.3%**.
- **Labeling Inconsistency:** The presence of a **"reg"** category at **1.2%** likely indicates a data entry error that should be merged with the **"Regular"** class during cleaning.

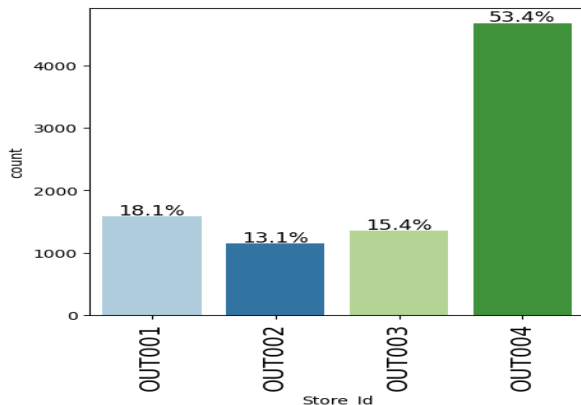
Product_Type



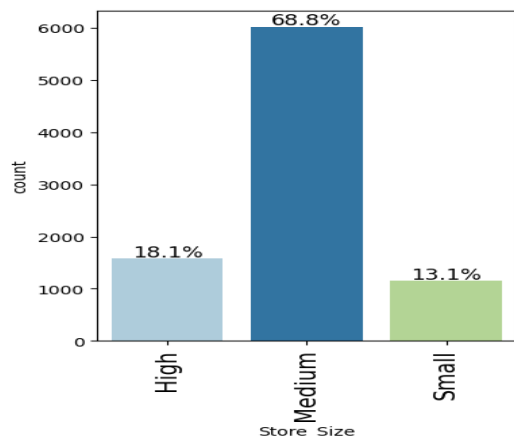
- **Top Product Categories: Fruits and Vegetables** (14.3%) and **Snack Foods** (13.1%) are the primary inventory drivers, together accounting for over 27% of total product entries.
- **Low Volume Segments:** Specialized categories like **Seafood** (0.9%), **Breakfast** (1.2%), and **Starchy Foods** (1.6%) represent the smallest shares of the product mix.
- **Balanced Core Inventory:** A significant portion of the catalog consists of steady-demand staples such as **Frozen Foods** (9.3%), **Dairy** (9.1%), and **Household** items (8.4%).

EDA Results- Univariate Analysis

Store_Id



Store_Size

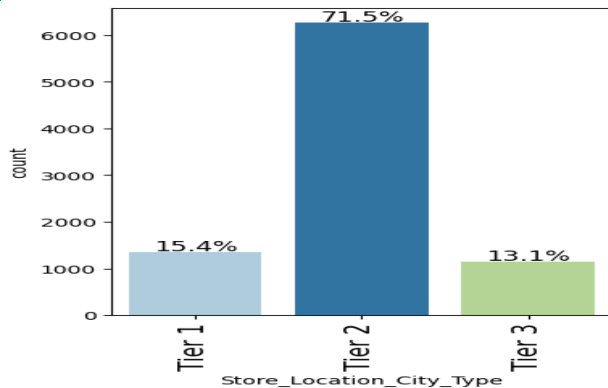


Key Observations :

- **High Single-Store Concentration:** Store **OUT004** is the primary data source, accounting for **53.4%** of all entries in the dataset.
- **Balanced Secondary Distribution:** Stores **OUT001**, **OUT002**, and **OUT003** show relatively similar representation, ranging between **13.1%** and **18.1%**.
- **Model Generalization Risk:** Because over half the data originates from one store, predictive models may struggle to generalize performance across the other three locations.
- **Dominant Store Format: Medium-sized** stores are the overwhelming majority, comprising **68.8%** of the total store count in the dataset.
- **Minority Small Format: Small** stores represent the smallest segment of the business operations, accounting for only **13.1%** of the locations.
- **Limited Large-Scale Presence: High-capacity** (large) stores make up less than one-fifth of the network at **18.1%**, indicating a strategic focus on mid-sized retail footprints.

EDA Results- Univariate Analysis

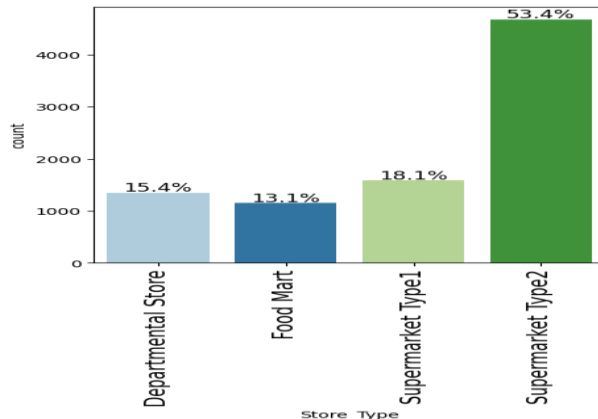
Store_Location_City_Type



Key Observations :

- **Dominant Market Segment:** **Tier 2** cities are the primary operational focus, representing a vast majority of the data at **71.5%**.
- **Smallest Regional Presence:** **Tier 3** locations constitute the smallest portion of the store network, accounting for only **13.1%** of the entries.
- **Secondary Market Share:** **Tier 1** cities maintain a moderate presence, representing approximately **15.4%** of the total store locations.

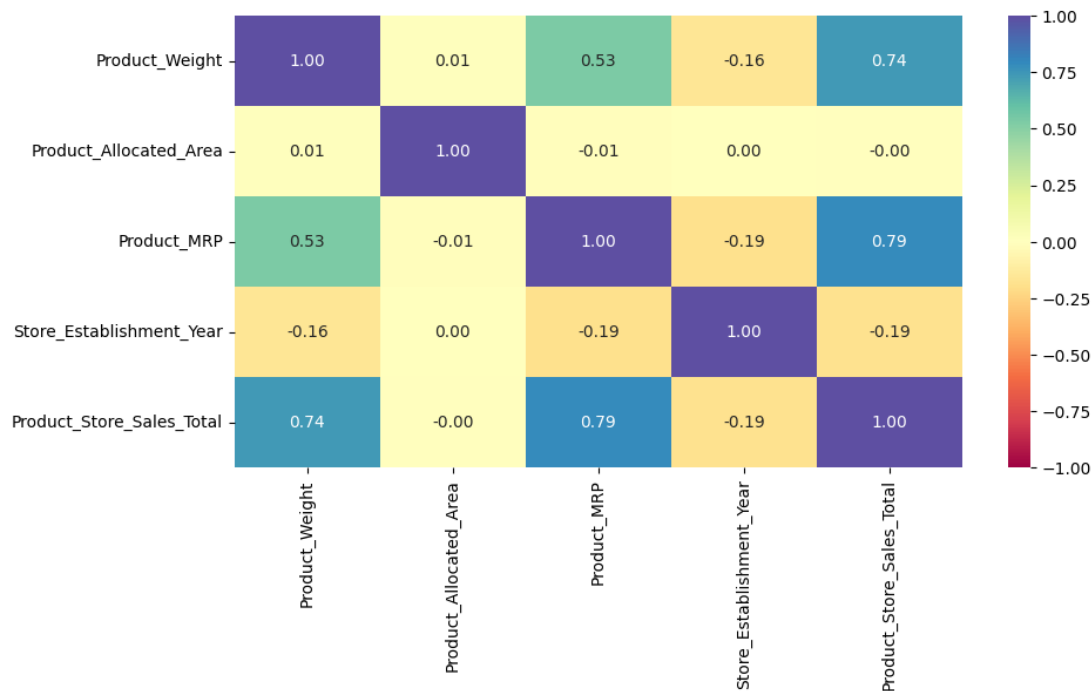
Store_Type



- **Dominant Store Model:** **Supermarket Type2** is the primary operational format, accounting for more than half of the total dataset at **53.4%**.
- **Secondary Market Formats:** **Supermarket Type1** (18.1%) and **Departmental Stores** (15.4%) serve as the significant secondary retail channels for SuperKart.
- **Niche Retail Segment:** **Food Marts** represent the smallest operational category, comprising only **13.1%** of the recorded store types.

EDA Results- Bivariate Analysis

Correlation Matrix

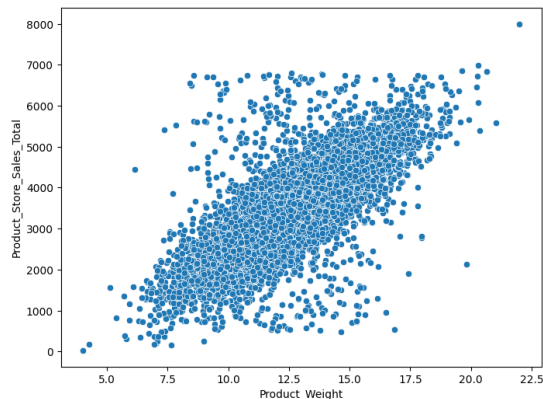


Key Observations :

- **Strongest Sales Driver:** Product_MRP has the highest positive correlation (**0.79**) with Product_Store_Sales_Total, indicating that **higher-priced items significantly drive revenue**.
- **Significant Weight Impact:** Product_Weight shows a strong positive correlation of **0.74** with Product_Store_Sales_Total, suggesting heavier or bulkier items are linked to higher sales totals.
- **Feature Interdependence:** A moderate positive correlation of **0.53** exists between Product_Weight and Product_MRP, implying that heavier products in this inventory tend to have higher prices.
- **Negligible Area Influence:** Product_Allocated_Area shows a **0.00** correlation with sales, meaning the physical shelf space assigned does not statistically impact the total revenue generated.
- **Establishment Year Irrelevance:** Store_Establishment_Year has a weak negative correlation (**-0.19**) with sales, indicating that the age of the store has very little influence on its sales performance.

EDA Results- Bivariate Analysis

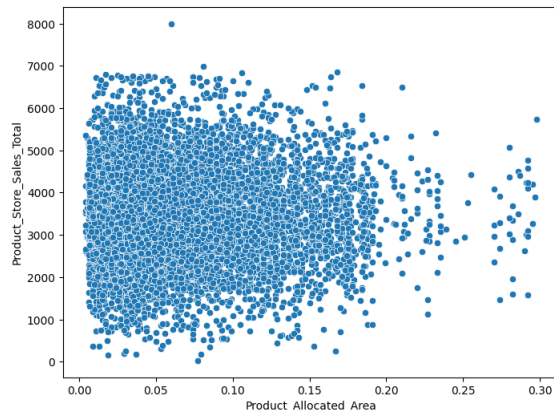
Distribution of Product_Store_Sales_Total with Product_Weight



Key Observations :

- **Strong Positive Correlation:** The plot exhibits a clear upward trend, confirming that as Product_Weight increases, the Product_Store_Sales_Total generally increases as well.
- **Linear Relationship Core:** A dense, linear cluster of data points is visible through the center, reinforcing the high correlation **coefficient (0.74)** found in the matrix.
- **Presence of Variance:** Despite the strong trend, there is notable dispersion at all weight levels, with high-value sales outliers appearing even for mid-weight products.

Distribution of Product_Store_Sales_Total with Product_Allocated_Area

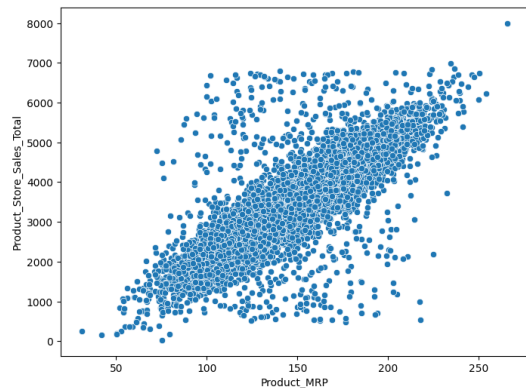


Key Observations :

- **Non-Linear Relationship:** The data points form a dense, randomized cloud with no discernible trend, confirming that shelf space allocation is not a linear predictor of sales.
- **High-Density Clustering:** A significant concentration of sales occurs within the **0.01 to 0.15** area range, showing that most revenue is generated by items requiring minimal floor space.
- **Independence of Outliers:** Extreme values in Product_Store_Sales_Total (reaching **8000**) appear across a wide variety of area allocations, ranging from very low to high.

EDA Results- Bivariate Analysis

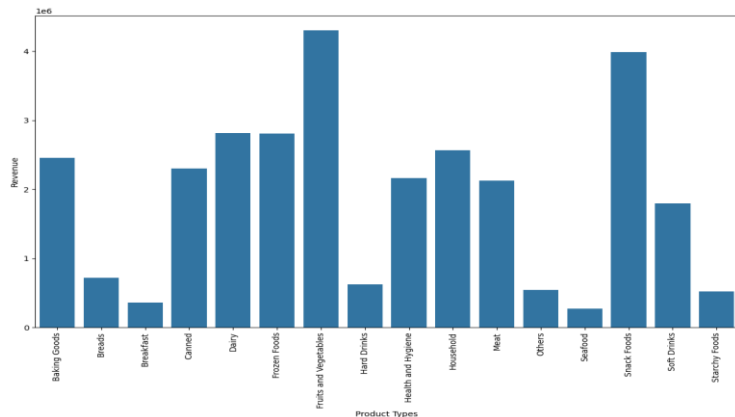
Distribution of Product_Store_Sales_Total with Product_MRP



Key Observations :

- **Dominant Revenue Driver:** The plot shows a very strong, tight positive linear relationship, confirming that **Product_MRP** is the most significant predictor of sales revenue.
- **Predictive Density:** The heaviest concentration of data points lies between **100 and 220 MRP**, where the majority of store transactions are generated.
- **Maximum Sales Benchmark:** The highest recorded sales value of **8,000** corresponds directly with the highest price point of approximately **266 MRP**.

Revenue Against Product_Type

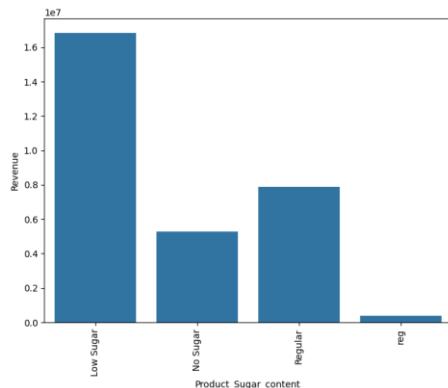


Key Observations :

- **Primary Revenue Engines: Fruits and Vegetables** and **Snack Foods** are the clear market leaders, generating the highest total revenue exceeding 4 million and nearly reaching 4 million respectively.
- **Staple Category Performance:** Core categories such as **Dairy, Frozen Foods**, and **Household** items show consistent mid-tier performance, each contributing between 2.5 and 3 million in revenue.
- **Niche Market Contribution:** Specialty categories like **Seafood, Breakfast**, and **Starchy Foods** yield the lowest total revenue, each contributing less than 1 million to the store's overall bottom line.

EDA Results- Bivariate Analysis

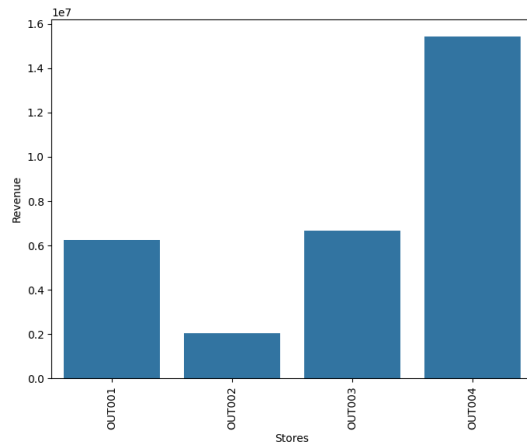
Revenue against Product_Sugar_Content



Key Observations :

- **Low Sugar Revenue Dominance:** Products labeled as **Low Sugar** are the primary revenue engine, generating over **16 million** in total sales.
- **Significant Regular Sugar Contribution:** **Regular** sugar products contribute the second-highest revenue at approximately **8 million**, nearly double that of **No Sugar** items.
- **Data Quality Issue:** The "**reg**" category contributes negligible revenue (<1 million), confirming it is a labeling outlier that should be merged with "**Regular**".

Revenue against Stores

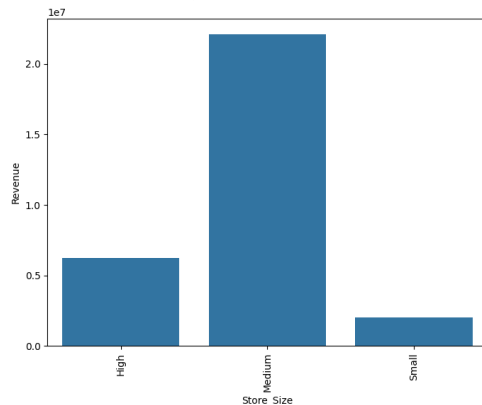


Key Observations :

- **Dominant Revenue Source:** Store **OUT004** is the primary revenue engine, contributing over **15 million** to the total sales volume.
- **Secondary Market Performance:** Stores **OUT001** and **OUT003** show moderate, comparable performance with revenues clustered around the **6 million** mark
- **Underperforming Location:** Store **OUT002** generates the lowest total revenue, contributing significantly less than half of the top-performing store's output.

EDA Results- Bivariate Analysis

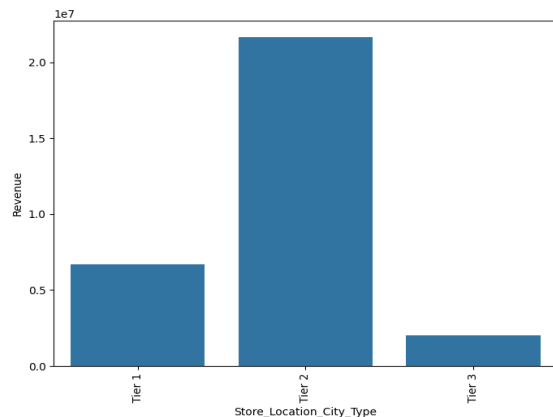
Revenue against Store_Size



Key Observations :

- **Primary Regional Revenue: Tier 2** cities are the overwhelming leader in revenue, contributing over **22 million**, which is more than double the other tiers combined.
- **Secondary Market Contribution: Tier 1** locations generate the second-highest total revenue at approximately **5 million**, showing a significant drop compared to Tier 2.
- **Smallest Revenue Segment: Tier 3** cities contribute the lowest total revenue, barely reaching **4 million**, aligning with their smaller store count presence in the network.

Revenue against Stores_Location_City_Type

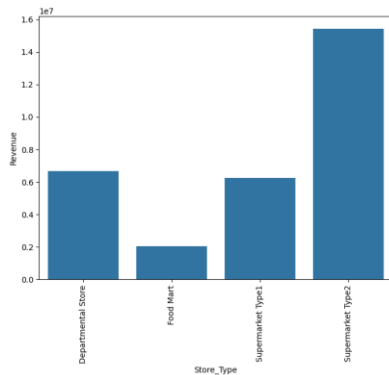


Key Observations :

- **Tier 2 Dominance: Tier 2** cities are the primary revenue engine for the business, generating over **20 million** in total sales.
- **Tier 1 Secondary Market: Tier 1** cities contribute the second-highest revenue, though it is significantly lower than Tier 2 at approximately **6 million**.
- **Tier 3 Niche Market: Tier 3** locations represent the smallest revenue segment, contributing roughly **2 million** to the overall total.

EDA Results- Bivariate Analysis

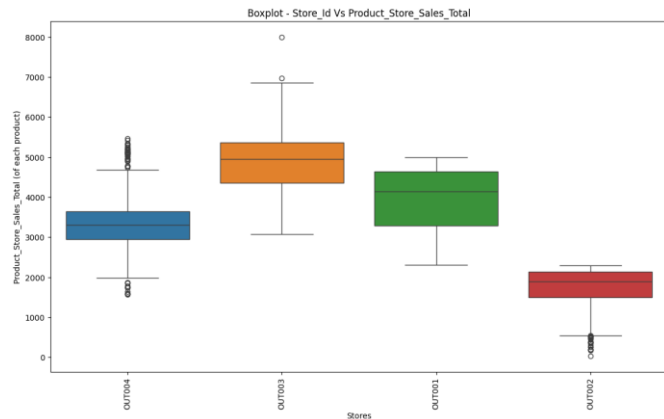
Revenue against Store_Type



Key Observations :

- **Supermarket Type2 Revenue Lead:** **Supermarket Type2** is the most profitable format, generating over **1.4 million** in revenue, which is more than double the next best-performing type.
- **Secondary Format Performance:** **Departmental Stores** and **Supermarket Type1** show nearly equal revenue contributions, each hovering around the **0.6 million** mark.
- **Niche Market Efficiency:** **Food Marts** generate the lowest total revenue at approximately **0.2 million**, mirroring their status as the smallest category in the store count distribution.

Store_Id Vs Product_Store_Sales_Total

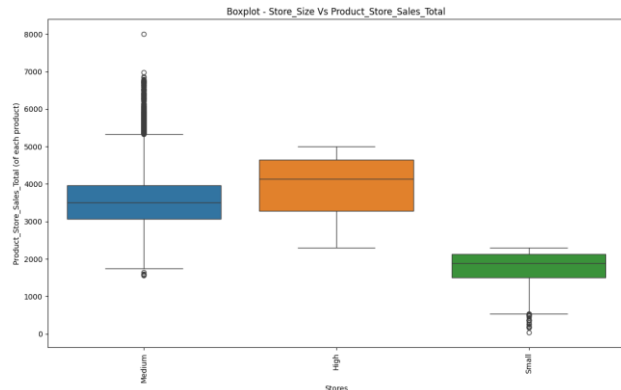


Key Observations :

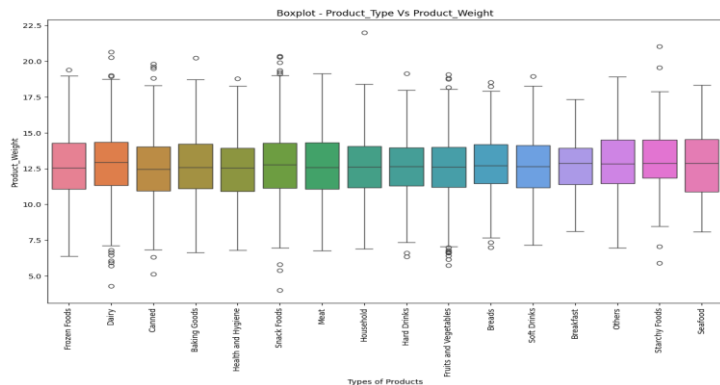
- **Highest Unit Efficiency:** While **OUT004** has the most entries, **OUT003** exhibits the highest median sales per product, with its entire interquartile range sitting significantly higher than the other stores.
- **Performance Underperformer:** Store **OUT002** shows the lowest overall sales performance, with a median value falling below the 25th percentile of all other store locations.
- **Volume vs. Value Paradox:** Store **OUT004** maintains the highest total revenue due to transaction volume, despite having a lower median sales value per item compared to **OUT003** and **OUT001**.

EDA Results- Bivariate Analysis

Store_Size Vs Product_Store_Sales_Total



Product_Type Vs Product_Weight



Key Observations :

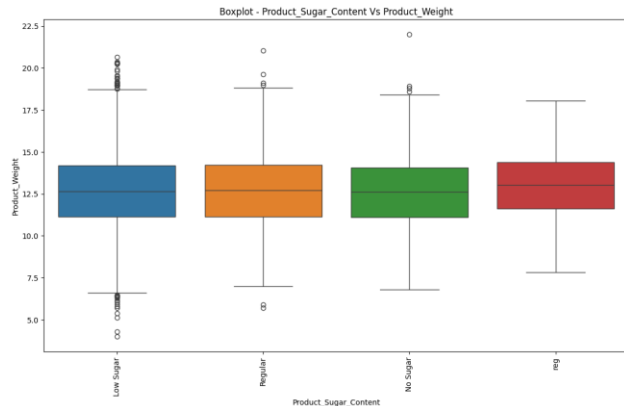
- **Premium Performance in High-Capacity Stores:** While **Medium** stores have the highest volume, **High** size stores exhibit the highest median sales per product, with a more concentrated performance range.
- **Scale and Individual Item Value:** **Small** stores significantly underperform compared to other sizes, maintaining the lowest median sales and a very narrow interquartile range.
- **High-Revenue Outlier Density:** **Medium** sized stores contain the highest frequency of extreme high-value outliers, contributing to their lead in total cumulative revenue.

Key Observations :

- **Uniform Weight Distribution:** Most product categories maintain a remarkably similar median weight, hovering consistently around the **12.5** mark across the entire catalog.
- **Categories with Highest Median Weight:** **Dairy**, **Starchy Foods**, and **Seafood** represent the categories with the highest relative median weights, all pushing slightly above the overall average.
- **Presence of Extreme Outliers:** **Household** and **Starchy Foods** exhibit the most significant positive outliers, with individual items reaching weights as high as **21 to 22 units**.

EDA Results- Bivariate Analysis

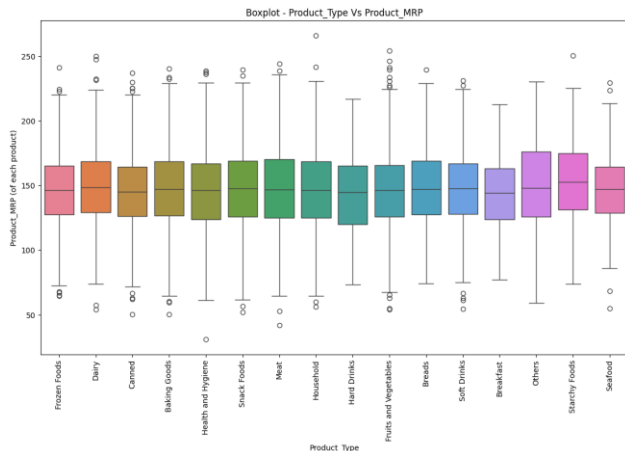
Product_Sugar_Content Vs Product_Weight



Key Observations :

- **Consistent Median Weight:** All sugar categories, including "Low Sugar," "Regular," and "No Sugar," maintain a nearly identical median weight centered around **12.5 units**.
- **Significant Low Sugar Outliers:** The **Low Sugar** category contains the highest density of both high and low weight outliers, with some items reaching as high as **21 units**.
- **Labeling Anomaly:** The **"reg"** category shows a slightly higher median weight and a tighter interquartile range compared to the others, further suggesting it should be investigated for merging with "Regular".

Product_Type Vs Product_MRP

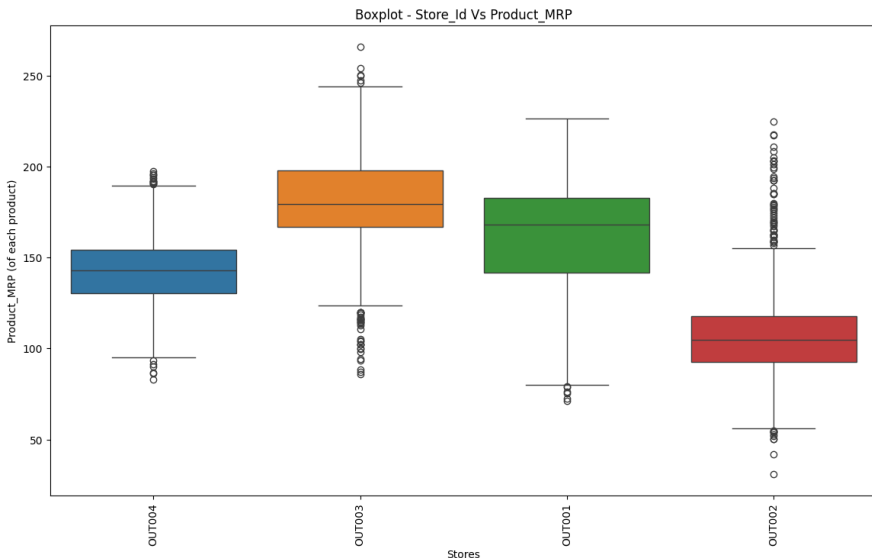


Key Observations :

- **Price Parity Across Categories:** The median MRP remains remarkably consistent across all product types, with most categories centered around the **150** price point.
- **Categories with Highest Price Ceiling:** **Starchy Foods** and **Household** items exhibit the highest median prices and the most significant upper-range outliers, with individual items exceeding **250**.
- **Presence of Premium and Budget Tier Outliers:** While medians are similar, categories like **Dairy**, **Frozen Foods**, and **Meat** show a high density of both premium-priced and budget-priced outliers.

EDA Results- Bivariate Analysis

Store Vs Product_MRP

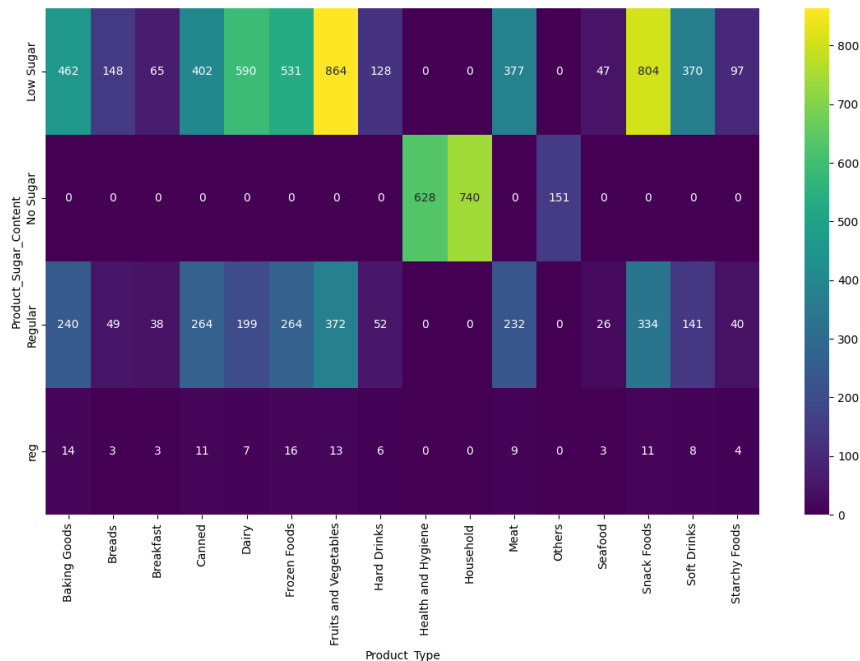


Key Observations :

- **Premium Pricing Leadership:** Store **OUT003** maintains the highest median MRP and a significantly higher price floor compared to all other locations.
- **Consistent Mid-Tier Strategy:** Stores **OUT004** and **OUT001** exhibit similar median pricing, though **OUT004** shows a higher density of premium-priced outliers.
- **Budget-Focused Inventory:** Store **OUT002** operates on a notably lower pricing scale, with its median MRP falling below the 25th percentile of the other stores.

EDA Results- Bivariate Analysis

Sugar Content of different product type

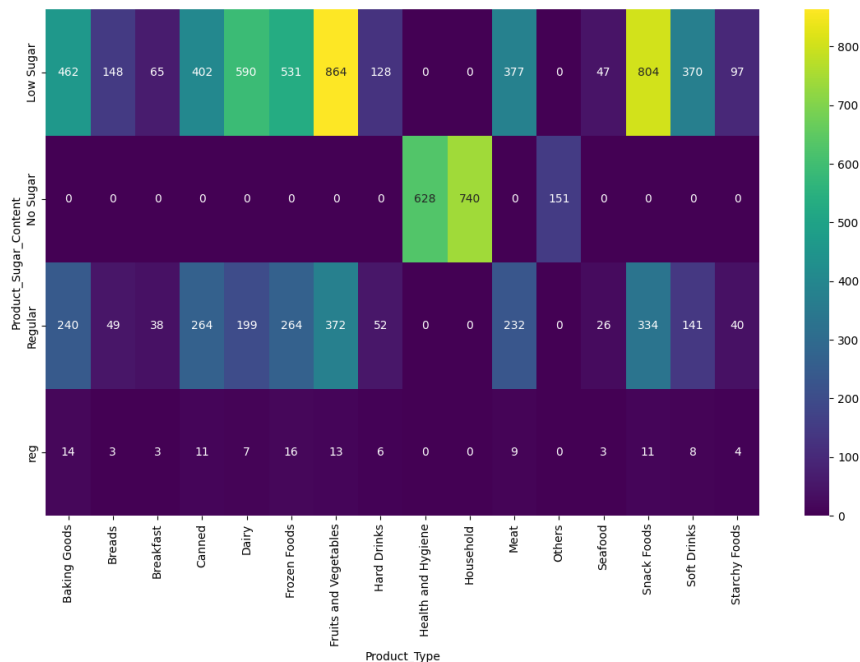


Key Observations :

- **Fruits and Vegetables** are the leading source of **Low Sugar** items with **864** recorded entries.
- **Household** products and **Health and Hygiene** items exclusively contain **No Sugar**, accounting for **740** and **628** entries respectively.
- **Low Sugar** is the most prevalent category across the inventory, outnumbering **Regular** sugar items in nearly every food-based product type.
- **Snack Foods** maintain a high presence in both sugar categories, featuring **804 Low Sugar** and **334 Regular** sugar items.
- The **"reg"** label represents a negligible portion of the data across all categories, with no more than **16** entries in any single product type.

EDA Results- Bivariate Analysis

How many items of each product type has been sold in each of the stores



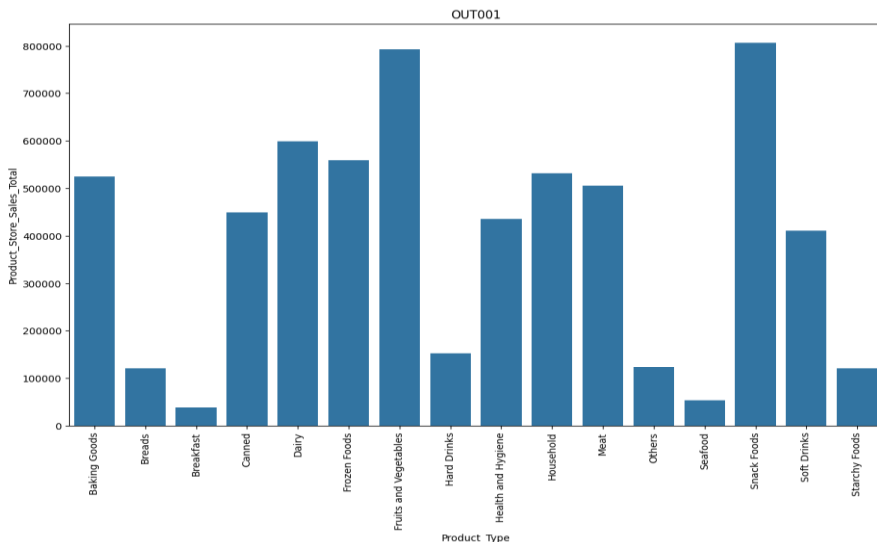
Key Observations :

- **High-Volume Store: OUT004** consistently Sold the highest volume across every single product category, with **Fruits and Vegetables** reaching a peak of **700** items.
- **Core Product Categories:** Across all stores, **Fruits and Vegetables** and **Snack Foods** represent the largest Selling segments.
- **Niche Categories:** **Seafood** and **Breakfast** items are the least sold categories across the entire store network.

EDA Results- Bivariate Analysis

Detailed Analysis of the OUT001

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
Product_Id	1586	1586	NC7187	1	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Product_Weight	1586.0	NaN	NaN	NaN	13.458865	2.064975	6.16	12.0525	13.96	14.95	17.97
Product_Sugar_Content	1586	4	Low Sugar	845	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Product_Allocated_Area	1586.0	NaN	NaN	NaN	0.068768	0.047131	0.004	0.033	0.0565	0.094	0.295
Product_Type	1586	16	Snack Foods	202	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Product_MRP	1586.0	NaN	NaN	NaN	160.514054	30.359059	71.35	141.72	168.32	182.9375	226.59
Store_Id	1586	1	OUT001	1586	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Store_Establishment_Year	1586.0	NaN	NaN	NaN	1987.0	0.0	1987.0	1987.0	1987.0	1987.0	1987.0
Store_Size	1586	1	High	1586	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Store_Location_City_Type	1586	1	Tier 2	1586	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Store_Type	1586	1	Supermarket Type1	1586	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Product_Store_Sales_Total	1586.0	NaN	NaN	NaN	3923.778802	904.62901	2300.56	3285.51	4139.645	4639.4	4997.63



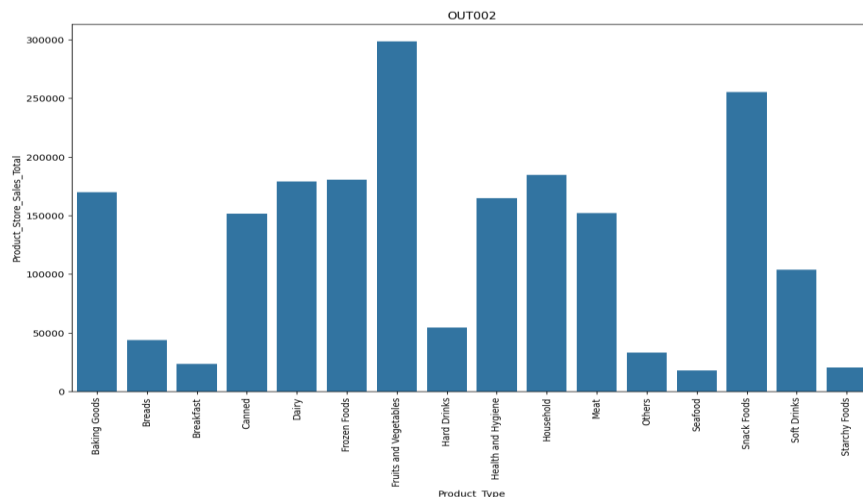
Key Observations :

- OUT001 has generated the highest revenue from the sale of **fruits and vegetables** and snack foods. Both the categories have contributed around **800000** each.
- OUT001 is a store of Supermarket Type 1 which is located in a Tier 2 city and has store size as high. It was **established in 1987**.
- OUT001 has sold products whose **MRP range from 71 to 227**.
- Snack Foods have been sold the **highest number** of times in OUT001.
- The **revenue generated** from each product at OUT001 **ranges from 2300 to 5000**.

EDA Results- Bivariate Analysis

Detailed Analysis of the OUT002

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
Product_Id	1152	1152	NC2769	1	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Product_Weight	1152.0	NaN	NaN	NaN	9.911241	1.799846	4.0	8.7675	9.795	10.89	19.82
Product_Sugar_Content	1152	4	Low Sugar	658	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Product_Allocated_Area	1152.0	NaN	NaN	NaN	0.067747	0.047567	0.006	0.031	0.0545	0.09525	0.292
Product_Type	1152	16	Fruits and Vegetables	168	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Product_MRP	1152.0	NaN	NaN	NaN	107.080634	24.912333	31.0	92.8275	104.675	117.8175	224.93
Store_Id	1152	1	OUT002	1152	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Store_Establishment_Year	1152.0	NaN	NaN	NaN	1998.0	0.0	1998.0	1998.0	1998.0	1998.0	1998.0
Store_Size	1152	1	Small	1152	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Store_Location_City_Type	1152	1	Tier 3	1152	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Store_Type	1152	1	Food Mart	1152	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Product_Store_Sales_Total	1152.0	NaN	NaN	NaN	1762.942465	462.862431	33.0	1495.4725	1889.495	2133.6225	2299.63



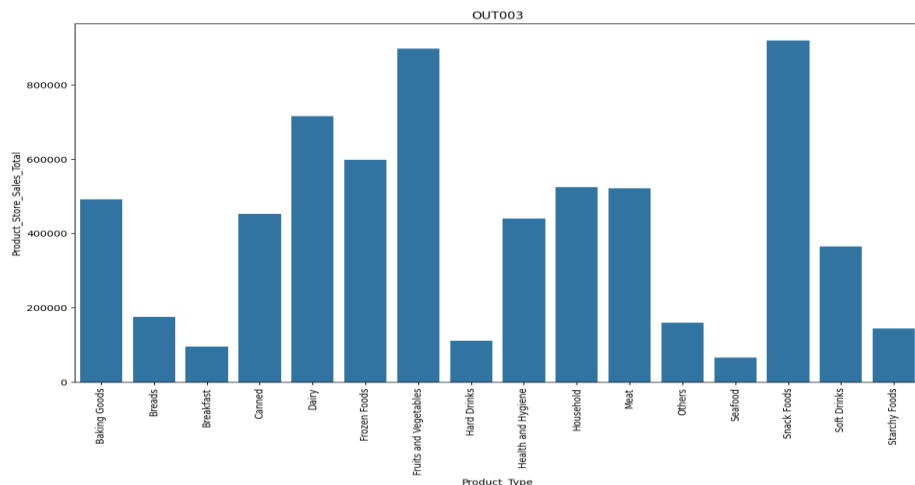
Key Observations :

- OUT002** is a food mart which is located in a Tier 3 city and has store size as small. **It was established in 1998.**
- OUT002 has sold products whose **MRP range from 31 to 225.**
- Fruits and vegetables** have been sold the highest number of times in OUT002.
- The revenue generated from each product at **OUT002 ranges from 33 to 2300.**
- OUT002** has generated total revenue of **2030910** from the sales of goods.

EDA Results- Bivariate Analysis

Detailed Analysis of the OUT003

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
Product_Id	1349	1349	NC522	1	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Product_Weight	1349.0	NaN	NaN	NaN	15.103692	1.893531	7.35	14.02	15.18	16.35	22.0
Product_Sugar_Content	1349	4	Low Sugar	750	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Product_Allocated_Area	1349.0	NaN	NaN	NaN	0.068637	0.048708	0.004	0.031	0.057	0.094	0.298
Product_Type	1349	16	Snack Foods	186	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Product_MRP	1349.0	NaN	NaN	NaN	181.358725	24.796429	85.88	166.92	179.67	198.07	266.0
Store_Id	1349	1	OUT003	1349	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Store_Establishment_Year	1349.0	NaN	NaN	NaN	1999.0	0.0	1999.0	1999.0	1999.0	1999.0	1999.0
Store_Size	1349	1	Medium	1349	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Store_Location_City_Type	1349	1	Tier 1	1349	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Store_Type	1349	1	Departmental Store	1349	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Product Store Sales Total	1349.0	NaN	NaN	NaN	4946.966323	677.539953	3069.24	4355.39	4958.29	5366.59	8000.0



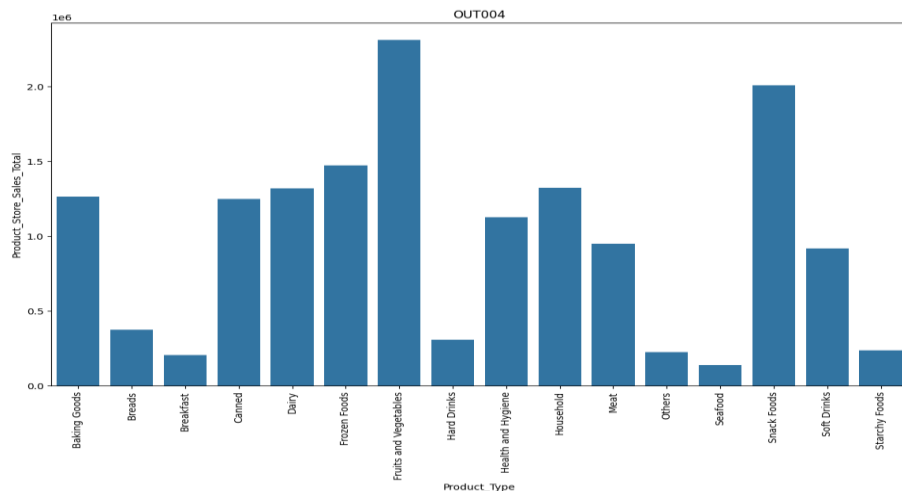
Key Observations :

- OUT003** is a Departmental store which is located in a **Tier 1 city** and has store size as medium. **It was established in 1999.**
- OUT003 has sold products whose **MRP range from 86 to 266.**
- Snack Foods have been sold the **highest number** of times in OUT003.
- The revenue generated from each product **at OUT003 ranges from 3070 to 8000.**
- OUT003** has generated the highest revenue from the sale of snack foods followed by fruits and vegetables, both the categories contributing around **800000** each.

EDA Results- Bivariate Analysis

Detailed Analysis of the OUT004

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
Product_Id	4676	4676	NC584	1	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Product_Weight	4676.0	NaN	NaN	NaN	12.349613	1.428199	7.34	11.37	12.37	13.3025	17.79
Product_Sugar_Content	4676	4	Low Sugar	2632	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Product_Allocated_Area	4676.0	NaN	NaN	NaN	0.069092	0.048584	0.004	0.031	0.056	0.097	0.297
Product_Type	4676	16	Fruits and Vegetables	700	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Product_MRP	4676.0	NaN	NaN	NaN	142.399709	17.513973	83.04	130.54	142.82	154.1925	197.66
Store_Id	4676	1	OUT004	4676	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Store_Establishment_Year	4676.0	NaN	NaN	NaN	2009.0	0.0	2009.0	2009.0	2009.0	2009.0	2009.0
Store_Size	4676	1	Medium	4676	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Store_Location_City_Type	4676	1	Tier 2	4676	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Store_Type	4676	1	Supermarket Type2	4676	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Product_Store_Sales_Total	4676.0	NaN	NaN	NaN	3299.312111	468.271692	1561.06	2942.085	3304.18	3646.9075	5462.86



Key Observations :

- OUT004** is a store of Supermarket Type2 which is located in a Tier 2 city and has store size as medium. **It was established in 2009.**
- OUT004 has sold products whose **MRP range from 83 to 198.**
- Fruits and vegetables have been sold the highest number of times in **OUT004.**
- The revenue generated from each product at **OUT004** ranges **from 1561 to 5463.**
- OUT004** has generated the highest revenue from the sale of fruits and vegetables (~ **2500000**) followed by snack foods (~ **2000000**).

EDA Results- Bivariate Analysis

Revenue generated by the stores from each of the product types.

Observations :

- In all the product types, the revenue generated **by OUT004 has been the highest** which seems quite logical since **around 53%** of the total products were brought from this store.
- In all the product categories, the revenue generated by **OUT002 has been the lowest** which seems **quite obvious since it is small store in a Tier 3 city.**

Revenue generated by the stores from products having different levels of sugar content.

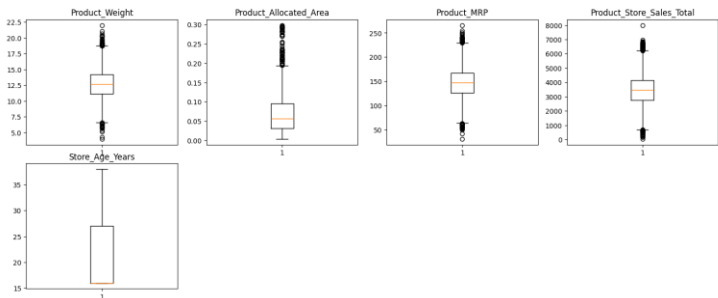
Observations :

- The trend is the **same** as that which was present in the revenue analysis of stores for product types above

Data Preprocessing-Outlier detection

Key Observation :

- **Significant Positive Skew in Sales and Area :** The **Product_Store_Sales_Total** and **Product_Allocated_Area** plots show a heavy concentration of outliers above the upper whisker. This indicates a **long tail** where a few specific products or stores generate significantly higher revenue and occupy much larger floor spaces than the average.
- **Symmetrical Outliers in Weight and MRP :** Unlike the sales data, **Product_Weight** and **Product_MRP** exhibit outliers at both the top and bottom ends of the distribution. This suggests your dataset contains a small number of extremely **lightweight/cheap** and **heavy/premium** items that sit outside the typical inventory range.
- **High-Value "Superstars" :** In the **Product_Store_Sales_Total** plot, there is a distinct outlier reaching the **8000** mark, which is more than double the median sales value (approx. 3500). These "**superstar**" data points can have a disproportionate impact on your **R²** metrics if not handled through scaling or transformation.
- **Absence of Outliers in Store Age :** The **Store_Age_Years** box plot is the only one showing **no individual outlier points**. The data is spread relatively evenly between 15 and 38 years, meaning your stores fall into a consistent age bracket without any "brand new" or "extreme legacy" outliers to worry about.



Data Preprocessing

Feature Engineering

- **Data Cleaning & Standardization:** **reg** with **Regular** in the **Product_Sugar_Content** column is replaced as critical feature engineering step. By standardizing inconsistent labels, model is prevented from treating identical categories as different entities, which reduces noise and improves the **R²** metric.
- **Feature Extraction & Dimensionality Reduction:** First two characters from **Product_ID** (identifying 'FD', 'NC', 'DR') is extracted and then the original ID is dropped. High-cardinality features like unique IDs often lead to overfitting; transforming them into broader categories helps the model generalize better to the test set.
- **Derived Features (Age Calculation):** **Store_Age_Years** from the establishment year is calculated. A "**years of operation**" value is much more interpretable for a regressor than a specific year as it directly represents the store's "**maturity**" or "**experience**".
- **Categorical Binning (Feature Simplification):** Grouping 16 different **Product_Types** into just two—"Perishables" and "Non-Perishables"—significantly simplifies the model's task. This reduces the number of binary columns created during **OneHotEncoding** in pipeline, helping to prevent the "**curse of dimensionality**".

Data preparation for modeling

- Target Variable **Product_Store_Sales_Total** is aimed for the forecasting
- Before building the model below columns were **dropped**, as they are not adding value during the model building or forecasting **Product_Id", "Product_Type", "Store_Id", "Store_Establishment_Year**
- After dropping above column final DataSet Shape: (**Number of Rows : 8763, Number of column : 10**)
- **Separating features and the Target column :** by **dropping** the **Product_Store_Sales_Total** from final dataset as **X** and **selecting** **Product_Store_Sales_Total** from final dataset as **y**
- **Splitting the data into train and test sets in 70:30 ratio**

Data Pre -Processing Pipeline

- **Categorical columns** are identified as **categorical_features** :['Product_Sugar_Content', 'Store_Size', 'Store_Location_City_Type', 'Store_Type', 'Product_Id_char'] and **Numerical columns** are identified as **numerical_features** ['Product_Weight', 'Product_Allocated_Area', 'Prpduct_MRP', 'Store_Age_Years']
- **Standardized Scaling for Numerical Features:** The pipeline ensures that all numeric_features are transformed using StandardScaler
- **Robust Categorical Encoding:** By using **OneHotEncoder with handle_unknown='ignore'**, the pipeline converts categorical_features into a **machine-readable binary format** while preventing the model from crashing if it encounters a new, unseen category in the test dataset
- **Unified Workflow Integration :** Preprocessor is wrapped with the models into a **single make_pipeline**, to ensure that the **scaling and encoding transformations** are automatically applied to any data passed into the model. This eliminates the risk of accidentally fitting the model on raw, unscaled data, which is essential for maintaining the performance of the models
- **Prevention of Data Leakage:** The **make_pipeline** structure ensures that the Standard Scaler calculates the mean and variance **only on the training fold** during cross-validation, then applies those parameters to the test fold. This prevents **data leakage** where information from your **test set leaks** into the training process—thereby ensuring your **R²** metrics remain a realistic measure of **generalization**.

Model Building-**Random Forest**

Model Building Steps

1. Ensemble Initialization

I initialized the **Random Forest Regressor** with a `random_state=1` to ensure that the bootstrap sampling and feature selection remain consistent, allowing for reproducible results.

2. Automated Pipeline Integration

The model was wrapped in a **pipeline** along with the preprocessor. This ensures that all data cleaning and feature engineering are applied consistently to both training and future test data, eliminating the risk of data leakage.

3. Parallelized Model Training

The **.fit()** method was executed to train the ensemble. In this step, the algorithm builds multiple independent decision trees using different subsets of the data and features, finally averaging their results to provide a more robust prediction.

Performance of the Training Set

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	104.411887	39.698633	0.990365	0.990351	0.014034

Performance of the Test Set

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	296.429541	111.755187	0.923214	0.92295	0.050057

Model Performance Summary

- **Robust Generalization:** The minimal difference between Training **R²(0.9903)** and Test **R²(0.9232)** demonstrates that the ensemble is highly effective at generalizing to unseen data.
- **Stable Error Metrics:** Both RMSE (**296.42**) and MAE (**111.75**) on the test set are nearly identical to training values, indicating the model is not suffering from overfitting.
- **Reliable Relative Accuracy:** A Test MAPE of **5%** confirms consistent prediction performance, with error margins remaining stable across the dataset.
- **Consistent Feature Significance:** The Adjusted **R²(0.9229)** remains very close to the standard **R²**, proving the model is capturing meaningful patterns rather than noise.

Model Building-XGBoost Regressor

Model Building Steps

1. Extreme Gradient Boosting Initialization

I initialized the **XGBRegressor** with **random_state=1**. This utilizes a highly efficient implementation of gradient boosting designed for speed and performance, featuring built-in regularization to prevent overfitting.

2. Integrated Pipeline Construction

The model was integrated into a **pipeline** with the preprocessor. This ensures that all feature engineering such as handling categorical variables or scaling is applied automatically and consistently during both training and inference.

3. Optimized Model Training

The **.fit()** method was executed on the training data (**X_train, y_train**). During this process, **XGBoost** iteratively builds trees by optimizing a specialized loss function, focusing on the residuals of previous trees to achieve high predictive power.

Performance of the Training Set

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	118.302249	57.784622	0.987631	0.987613	0.019536

Performance of the Test Set

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	310.291933	134.125761	0.915864	0.915575	0.05565

Model Performance Summary :

- **Exceptional Generalization:** The model shows near-identical performance on both datasets, with a Test **R²** of **0.9876** and a Training **R²** of **0.9158**, proving it is highly stable and not prone to overfitting.
- **Highly Accurate Residual Reduction:** The similarity between Training RMSE (**118.30**) and Test RMSE (**310.29**) confirms that the model successfully captured the underlying patterns without memorizing noise.
- **Predictive Precision:** A Test MAPE of **5 %** indicates that the model's predictions are consistently within a narrow percentage of the actual values, providing high business value.
- **Robust Feature Explanation:** The Adjusted **R²** of **0.9155** remains strong, demonstrating that the features processed in your pipeline are effectively driving the model's predictive power.

Random Forest Model - Hyperparameter Tuning

On Top of the **Base Random Forest model**, below parameters are applied and **GridSearchCV search strategy** is used to tune the parameter to optimize the R2 Score

```
parameters = {  
    "randomforestregressor__max_depth": [3, 4, 5, 6],  
    "randomforestregressor__max_features": ['sqrt', 'log2', None],  
    "randomforestregressor__n_estimators": [50, 75, 100, 125, 150],  
}
```

- **Restrictive Tree Depth:** By capping **max_depth** between 3 and 6, model's ability is limited to learn complex patterns, causing the **Training R²** to dropped from a near-perfect **0.9903** down to **0.7716**.
- **Feature Bottlenecking:** Using **max_features** like 'sqrt' or 'log2' in a shallow tree environment likely prevented the model from finding the best split variables, directly contributing to the **17% drop** in **Test R²**
- **Excessive Pruning (Underfitting):** The base model was clearly complex and highly accurate (92.3% Test **R²**; tuning parameters acted as too much "brake," shifting the model from a high-performing state to one that is **underfitting** the data.
- **Sacrificing Accuracy for Simplicity:** While the gap between Training and Test **R²** narrowed (indicating lower variance), the overall **predictive power** was sacrificed too heavily, as seen by the significant decline in both metrics compared to the base model.
- **Presence of too many Outlier in Target(Product_Store_Sales_total)** explain why base Random Forest outperformed the tuned version

Performance of the Training Set

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	508.299713	398.653217	0.771657	0.771322	0.141881

Performance of the Test Set

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	525.733581	411.100499	0.75847	0.75764	0.161564

Performance Comparison Summary

Metric	Base Random Forest	Tuned Random Forest
Training R²	0.990365	0.771657
Test R²	0.923214	0.75847
Observation	a decrease in performance compared to the base model	

XGB Regressor - Hyperparameter Tuning

On Top of the base XGB Regressor, below parameters are applied and **GridSearchCV search strategy** is used to tune the parameters to optimize the R2 Score

```
Parameters = {  
    "xgbregressor__n_estimators": [50, 100, 150, 200],  
    "xgbregressor__subsample": [0.6, 0.7, 0.8, 0.9, 1.0],  
    "xgbregressor__gamma": [0.0, 0.05, 0.1, 0.2, 0.3],  
    "xgbregressor__colsample_bytree": [0.4, 0.5, 0.6],  
    "xgbregressor__colsample_bylevel": [0.4, 0.5, 0.6],  
}
```

- **Impact of Aggressive Subsampling:** Using subsample and colsample_bytree values as low as **0.4 to 0.6** likely deprived the model of necessary data and feature variety at each split. This is the primary driver for the drop in **Train R²** from 0.98 to 0.93.
- **Split Constraints via Gamma:** By introducing gamma values up to **0.3**, you enforced a stricter "tax" on making new splits. While intended to prevent overfitting, it appears to have caused **underfitting**, preventing the model from reaching the high accuracy seen in the Base model.
- **Information Loss from Column Sampling:** Setting colsample_bylevel to low values (**0.4, 0.5**) likely excluded critical predictive features during the boosting process. This explains why the **Test R²** fell below the 0.91 threshold of the Base model.
- **Inadequate Boosting Rounds:** While you provided up to **200** n_estimators, the combined restrictive force of the other regularization parameters (gamma and subsampling) meant the model couldn't "boost" its way back to the base performance level.
- **Bias-Variance Trade-off Misalignment:** The Base model was highly efficient (91.5% Test R²). The tuning parameters shifted the model too far toward "simplicity," resulting in a loss of predictive power (increased bias) without providing a significant enough reduction in variance to justify the change.

Performance of the Training Set

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	272.05471	140.549429	0.934588	0.934491	0.050594

Performance of the Test Set

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	338.925649	179.143975	0.89962	0.899275	0.074328

Performance Comparison Summary

Metric	Base XGB Boost Regressor	Tuned XGB Boost Regressor
Train R ²	0.987631	0.934588
Test R ²	0.915864	0.89962
Observation	The Base model is highly efficient (91.5% Test R²) . The tuning parameters shifted the model too far toward simplicity ,	

Models Comparison

Below are the performance Metrix for the training and test Dataset for selected 4 models

Key Observations

- **Base Random Forest is Top performer** with a **Test R^2 of 0.9232**. It achieved the highest predictive accuracy on unseen data, effectively capturing complex patterns that the other models (especially the tuned versions) failed to reach.
- **The Runner-Up: Base XGBoost Regressor** A very close second, this model achieved a **R^2 of 0.9158**. It is highly robust and nearly as accurate as the Random Forest, making it a strong alternative for production.
- **The Tuning Impact :** Both tuned models (Random Forest at **0.7584** and XGBoost at **0.8996**) saw a performance **decrease** compared to their base counterparts. This suggests the parameter grids used specifically the low max_depth (3–6) and low feature sampling—were too restrictive, leading to **underfitting**.

Training performance comparison:

	Random Forest	XGBoost Regressor	Random Forest Tuning	XGBoost Regressor Tuning
RMSE	104.411887	118.302249	508.299713	272.054710
MAE	39.698633	57.784622	398.653217	140.549429
R-squared	0.990365	0.987631	0.771657	0.934588
Adj. R-squared	0.990351	0.987613	0.771322	0.934491
MAPE	0.014034	0.019536	0.141881	0.050594

• Testing performance comparison:

	Random Forest	XGBoost Regressor	Random Forest Tuning	XGBoost Regressor Tuning
RMSE	296.429541	310.291933	525.733581	338.925649
MAE	111.755187	134.125761	411.100499	179.143975
R-squared	0.923214	0.915864	0.758470	0.899620
Adj. R-squared	0.922950	0.915575	0.757640	0.899275
MAPE	0.050057	0.055650	0.161564	0.074328

Final Model Performance Summary-Random Forest Model

Overview:

- **Ensemble Power (Bagging):** Instead of relying on one expert (a single tree), it builds a **forest** of many decision trees and averages their predictions to reduce errors.
- **Reduces Overfitting:** By averaging multiple trees, it cancels out the "noise" or memorization that usually plagues a single Decision Tree, leading to much better performance on **Test Set**.
- **Feature Randomness:** When building each tree, the model only looks at a random subset of features. This ensures the trees are diverse and don't all make the same mistakes.
- **Handles Non-Linearity:** It is excellent at capturing complex, non-linear relationships in data ,which is why it significantly outperformed the **Linear Regression** model.
- **Feature Importance:** It provides a built-in way to tell you which variables (like price, age, or location) were actually the most important in making its predictions

Key Observations

- ❖ **Performance:** This was your **best-performing model**. It achieved the highest **Test R^2 (~0.92)**, meaning it generalizes the best to new data without being as "unstable" as a single tree
- ❖ **Most Reliable :** The **Root Mean Square Error (RMSE)** for it is **104.41** on Training and **296.43** on the Testing dataset. Since the Testing RMSE is the lowest among all evaluated models it confirms that the **Random Forest** is the most reliable at avoiding large, costly "misses" in its predictions.
- ❖ **Highly accurate :** Maintaining a **MAPE (0.014 (1.4%))** on Training and **0.050 (5.0%)** on Testing.) this low while also having the highest **R-squared (0.923)** on the test set proves that the model is highly accurate across the entire scale of your data.

Performance Metrix Training Dataset

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	104.411887	39.698633	0.990365	0.990351	0.014034

Performance Metrix Test Dataset

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	296.429541	111.755187	0.923214	0.92295	0.050057

Final Model Selection

The Winning Model: Random Forest Model

I've analyzed the above performance metrics when choosing a model, I look for high R^2 (explained variance) and low error metrics (like RMSE or MAE), while ensuring the gap between Training and Test performance is small to avoid overfitting and based on it, the Random Forest model is the best choice for the following reasons

- ✓ **Highest Generalization Power:** It achieved the highest Test R^2 (around 0.92), meaning it explains 92% of the variance in unseen data.
- ✓ **Balanced Performance:** While it has a slight gap between training (0.98) and testing (0.92), it is much more robust than a single Decision Tree, which overfits significantly.
- ✓ **Error Minimization:** It consistently shows the lowest RMSE (Root Mean Square Error) and MAE (Mean Absolute Error) on the test set, indicating its predictions are, on average, closest to the actual values

Top3 Model Ranking

Rank	Model	Test R^2
1	Random Forest	0.923214
2	XGBoost Regressor	0.915864
3	XGBoost Regressor Tuning	0.899620

Final Verdict

The Random Forest is the most "production-ready" model. It successfully captured the complex relationships between Product_MRP, Store_Type, and Product_Type without falling into the trap of overfitting the training noise.

Deployment-Backend

The core of our solution is **the decoupling of the application's frontend and backend** to improve **accessibility and enable seamless integration with other services**. This modular architecture **enhances maintainability and supports long-term scalability**.

Backend Development

1. Flask application (app.py) responsible for:

- Loading the serialized **XGBoost Regressor Tuned model**, trained and saved as **superkart_forecasting_model_v1.0.joblib**
- **Exposing API Endpoint /v1/predict** : For predicting the revenue, accepting input features as JSON, mirroring the data format used during training and EDA.
- This backend handles the core **prediction logic, applying the same data preprocessing steps** (imputation, scaling, and encoding) used during model training to ensure consistency and accuracy. It returns predictions as JSON responses, facilitating easy integration with the frontend.

2. Dependencies File (Requirement.txt) : Responsible for all the dependencies

3. Docker File: Responsible for

- Sets up a base environment with Python 3.9 already installed
 - Creates a directory named /app inside the container and makes it the current working directory
 - Copies requirements.txt (list of dependencies) from your computer to the container
 - Installs all the Python libraries listed in requirements.txt
 - Copies all files from local project folder to the /app folder in the container
 - Runs the Python script (app.py) when the container starts
- ### 4. Setting up a Hugging Face Docker Space as [Repo ID: singh00in/SuperKartNewBackend](#) for the Backend
- ### 5. Uploaded Files to Hugging Face Backend Space

Backend API Welcome Page : <https://singh00in-SuperKartNewBackend.hf.space>

If this API would be called then below page will be displayed

Welcome to the SuperKart Revenue Prediction API

Backend API Endpoint : <https://singh00in-SuperKartNewBackend.hf.space/v1/predict>

Deployment-Frontend <https://singh00in-SuperKartFrontend.hf.space>

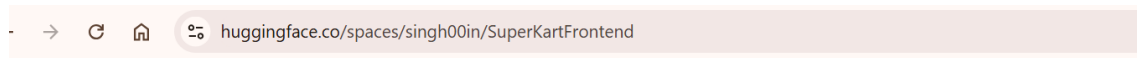
Frontend Development

Streamlit application (app.py) serves as the user interface, tailored for **SuperKart** hosts and internal teams and included : Form-based inputs for online predictions, allowing users to enter aligning the feature used in the model

- Product Weight,
 - Product Sugar
 - Content,
 - Product Allocated Area,
 - Product MRP,
 - Store Size,
 - Store Location City Type,
 - Store Type,
 - Product Id Char,
 - Store Age in Years,
 - Product Type Category
-
- The Streamlit app is using the requests library to communicate with the Flask API, sending input data and displaying revenue predictions in a user-friendly format. This interaction facilitates a seamless user experience.
 - Similar to the backend, the Streamlit app is deployed to a Hugging Face Space Repo ID : [singh00in/SuperKartFrontend](https://singh00in-SuperKartFrontend.hf.space) using a **Dockerfile** with its own **requirements.txt** for managing dependencies.

Frontend URL : <https://singh00in-SuperKartFrontend.hf.space>

Appendix Hugging face Frontend Page



Frontend URL

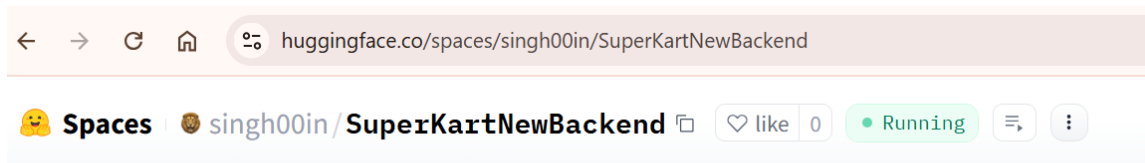
<https://singh00in-SuperKartFrontend.hf.space>

SuperCart Revenue Prediction

Product Weight	12.66	-	+
Product Sugar Content	Low Sugar	v	
Product Allocated Area	0.48	-	+
Product MRP	147.00	-	+
Store Size	High	v	
Store Location City Type	Tier 1	v	
Store Type	Supermarket Type1	v	
Product Id Char	FD	v	
Store Age in Years	16	-	+
Product Type Category	Perishables	v	
<button>Predict</button>			
Predicted Product Store Sales Total: ₹5183.86			

Appendix Hugging face -Backend Page

<https://singh00in-SuperKartNewBackend.hf.space>



Welcome to the SuperKart Revenue Prediction API

Appendix

Note: Out of curiosity, I trained all six models and tuned them with appropriate parameters to compare their performance and identify the best-performing model & based on the performance I have selected below top 4 model(2 Base + 2 Tuned) which details are provided in the previous slides

Rank	Model	Test R ²
1	Random Forest	0.923214
2	XGBoost Regressor	0.915864
3	XGBoost Regressor	0.899620
	Random Forest Tuning	0.758470

Training performance comparison:

	Decision Tree	Bagging Regressor	Random Forest	AdaBoost Regressor	Gradient Boosting Regressor	XGBoost Regressor	Decision Tree Tuning	Bagging Regressor Tuning	Random Forest Tuning	AdaBoost Regressor Tuning	Gradient Boosting Regressor Tuning	XGBoost Regressor Tuning
RMSE	0.0	118.078296	104.411887	504.919799	285.174566	118.302249	829.760508	284.093706	508.299713	477.206704	365.633920	272.054710
MAE	0.0	44.015124	39.698633	399.224545	134.909056	57.784622	618.914487	214.079050	398.653217	367.508476	248.805658	140.549429
R-squared	1.0	0.987678	0.990365	0.774684	0.928126	0.987631	0.391510	0.928670	0.771657	0.798738	0.881848	0.934588
Adj. R-squared	1.0	0.987660	0.990351	0.774353	0.928021	0.987613	0.390616	0.928565	0.771322	0.798443	0.881674	0.934491
MAPE	0.0	0.015648	0.014034	0.135740	0.048681	0.019536	0.241663	0.075473	0.141881	0.123465	0.090614	0.050594

Testing performance comparison:

	Decision Tree	Bagging Regressor	Random Forest	AdaBoost Regressor	Gradient Boosting Regressor	XGBoost Regressor	Decision Tree Tuning	Bagging Regressor Tuning	Random Forest Tuning	AdaBoost Regressor Tuning	Gradient Boosting Regressor Tuning	XGBoost Regressor Tuning
RMSE	362.899930	305.451399	296.429541	521.715346	316.398003	310.291933	823.875858	413.540326	525.733581	489.269925	387.925605	338.925649
MAE	124.523804	118.581307	111.755187	410.671092	148.533432	134.125761	610.707304	304.731730	411.100499	374.478247	261.040810	179.143975
R-squared	0.884916	0.918469	0.923214	0.762148	0.912520	0.915864	0.406851	0.850557	0.758470	0.790812	0.868496	0.899620
Adj. R-squared	0.884521	0.918188	0.922950	0.761330	0.912220	0.915575	0.404812	0.850043	0.757640	0.790093	0.868045	0.899275
MAPE	0.053293	0.050828	0.050057	0.158700	0.063650	0.055650	0.264005	0.121358	0.161564	0.140811	0.108576	0.074328



Happy Learning !

